rogii-wellbore-geology-prediction

18th Place Solution

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I would like to express my gratitude to the competition host and the Kaggle staff for organizing this outstanding competition.

I would also like to thank Tucker @tuckerarrants . Initially, I used a large-scale stacking architecture that combined PF, statistical features from neighboring wells, and features extracted by 1D and 2D CNNs. After seeing Tucker's comment that a single model using only data from the target well could achieve a CV score of 5.x, I was encouraged to shift my focus toward improving a single-model solution.

TL;DR

- We **turn geosteering into an image-recognition task**: the matching surface between the lateral GR and the typewell GR is rendered as a 2D image, and a CNN picks the correct path (ridge) through it.
- The vertical axis of the image is defined as the **offset from a reference line (chord)** drawn from the known well trajectory. This shrinks the range of values to predict, so the whole well fits in a **narrow vertical window at high resolution** — the single biggest lever.
- With only 773 training wells, we diversify training data with **synthetic samples** (typewell depth shifts / geology swaps) rendered with a measured noise model.
- The final submission is a **fixed-weight linear blend** of chord models with different backbones and resolutions.

Problem intuition

Each well has ground-truth TVT only for its first part (the prefix); the rest must be predicted. Since the geology is layered, "which depth of the typewell does the current GR look like?" is the evidence for TVT — this is geosteering.

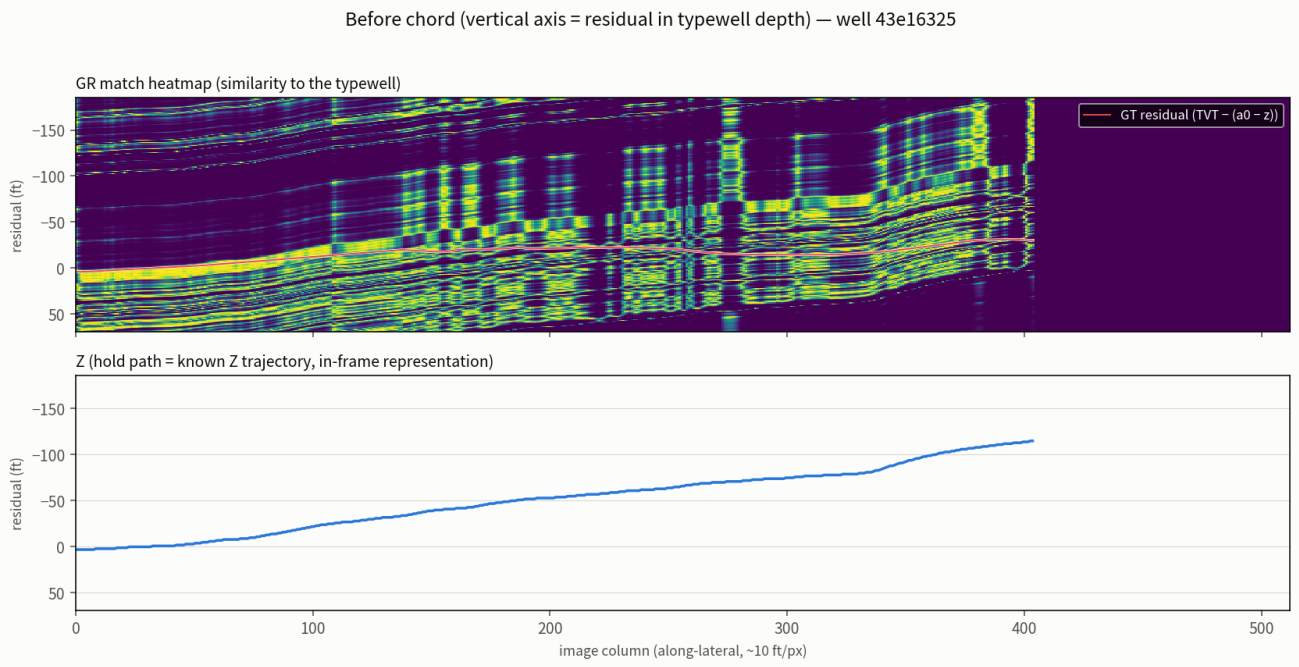
The difficulty is the **multimodality of the matching**. GR columns contain many self-similar intervals, so several candidate depths line up as parallel ridges. Most of the error comes from a small number of wells that **jump onto the wrong ridge** (wrong-mode), so the real problem is choosing the right ridge, not per-step precision.

Method: render each well as one image and estimate the path

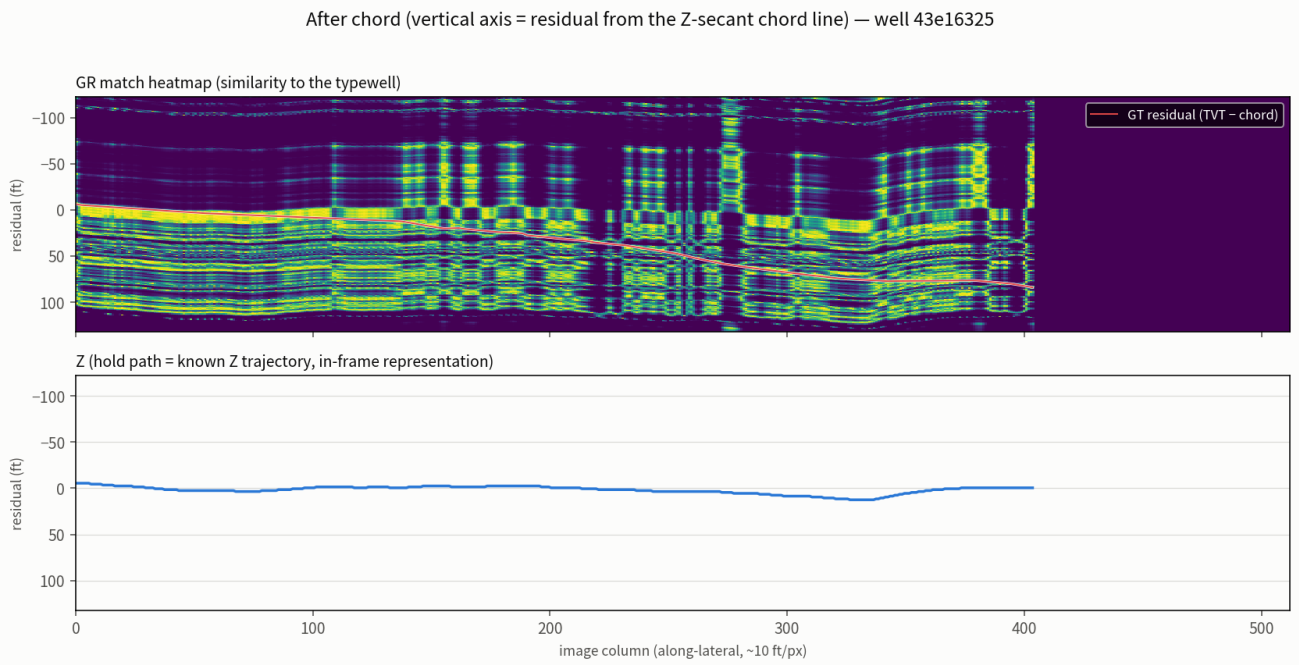
Instead of regressing row by row, we convert the entire well into a single 2D image and solve "trace the correct path across the image".

Vertical axis as the residual from a reference line (chord) — the core idea

The horizontal axis is the position along the well; the vertical axis is the candidate TVT. The goal here is to **narrow the range of values to predict and spend the savings on vertical resolution**. If the vertical axis is naively the typewell depth, the true path drifts widely across the image, so covering every well requires a tall window (= coarse resolution or a huge image).



Instead, we draw a straight line connecting the start and end of the known well trajectory (Z) — the **chord** — and use the **offset from this line** as the vertical axis. The range to predict shrinks dramatically:

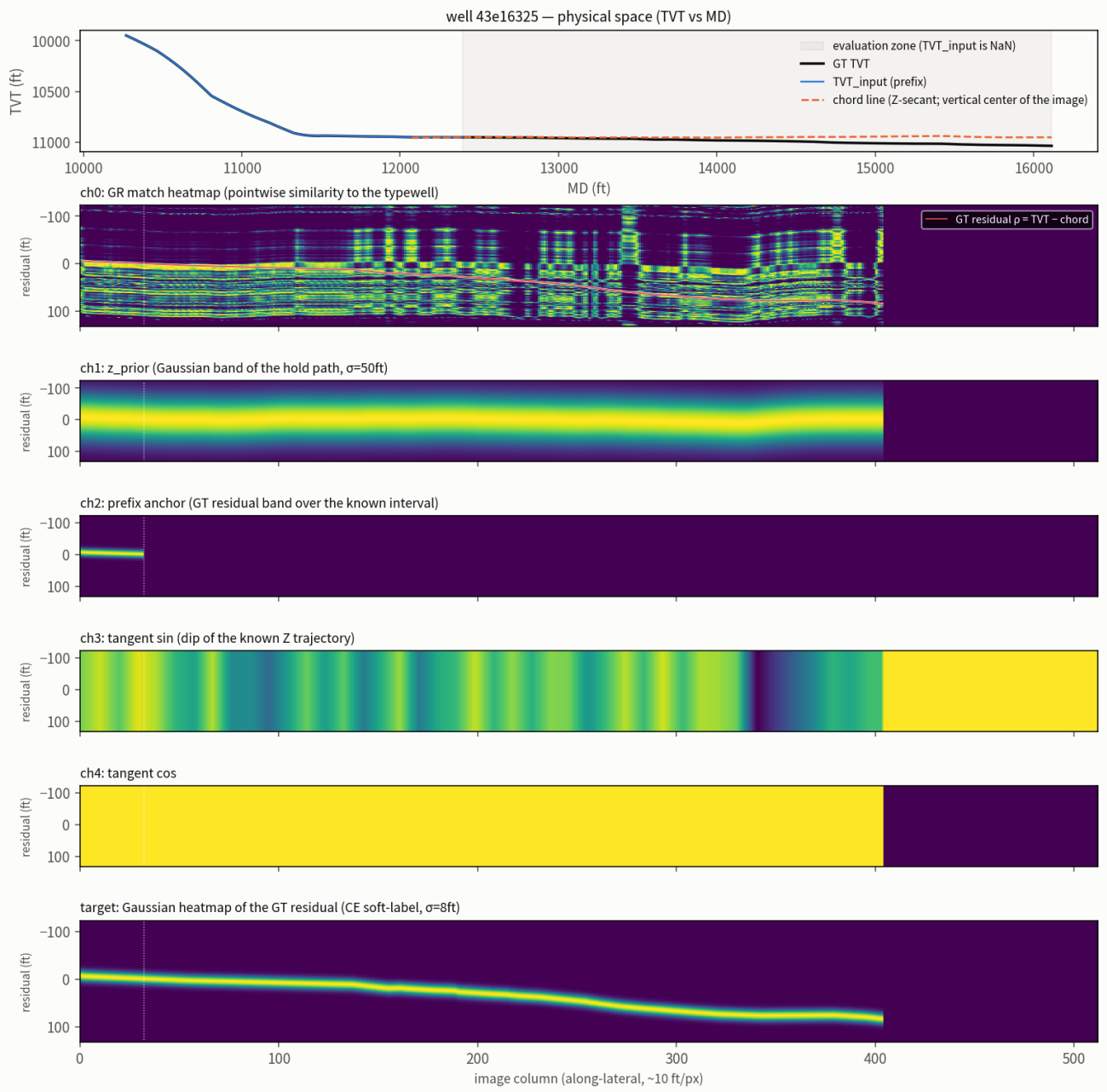


- With the same image size, each vertical pixel can be made finer — down to 0.5–1 ft/bin.
- The GR matching scores (next section) are precomputed once on the typewell depth axis and simply re-read with a per-column vertical shift along the chord (no recomputation). `TVT = chord + residual` always restores the physical value.
- We also tried a two-stage setup that draws the reference line from a first-stage model's predictions, but once resolution and window size could be scaled directly, **the simple chord drawn mechanically from the known trajectory was the strongest**.

Input image

The horizontal axis is the position along the well (~10 ft/px, 512 columns); the vertical axis is the offset from the chord (0.5–1 ft/bin and a ±96–128 ft window depending on the model). Five channels:

- **GR matching score** — how well the GR at each position matches the typewell GR at each candidate depth (precomputed). The true path and all the misleading false paths are visible here as multiple ridges.
- **Prior from the known trajectory** — the path assuming TVT stays at its last known value, drawn as a soft band. A cue for the absolute depth level.
- **Ground truth over the known interval** — the true path in the prefix, where the answer is given. Marks "start from here".
- **Trajectory dip (sin/cos)** — the local inclination of the known well trajectory.



Model

The model takes this image and estimates, for each column, where the true path lies vertically. We used two head types.

The **heatmap head** is a UNet that outputs a per-column probability distribution (heatmap) over vertical positions. It is trained with cross entropy against a Gaussian soft label centered at the true position ($\sigma = 8$ ft); at inference the expectation of the heatmap is added back to the chord to restore TVT (the bottom panel of the figure above is the training target).

The **detection head** follows an object-detection-style formulation: the vertical axis is divided into coarse cells, and for each column the model classifies which cell contains the true path and regresses the offset from the cell center. Combined with a DINOv3-pretrained ViT backbone, it served as an ensemble member whose errors correlate weakly with the heatmap models.

Synthetic data

A real lateral samples only ~30 ft of the typewell column. During training:

- **shift**: pretend the well was drilled s0 ft deeper/shallower in its typewell and re-render the GR from the typewell (geometry and ground truth stay real; only the evidence channel is diversified).
- **swap**: additionally replace the GR waveform (the geology) with another well's typewell.
- Re-rendered GR gets a **measured noise model** (per-well gain + colored noise, calibrated to the real coherence). Too-clean synthetic GR was actively harmful.
- **Dose decay** — linearly annealing the synthetic fraction to zero late in training — also helped for some models.

Augmentation

- **crop** — jitter the vertical window center and randomize the known/unknown boundary (prediction start), exposing the model to various prefix lengths and window positions.
- **vertical flip** — the residual from the chord is symmetric under sign flip, so flip the input and target together ($p = 0.5$). The same symmetry is used at inference and for checkpoint selection by averaging with the flipped view (vflip TTA).
- **horizontal flip** — reverse the drilling direction (flipping only the horizontal section worked best).
- **horizontal resize** — random horizontal resize with $U(0.7, 1.3)$. At inference, long wells are compressed to fit the 512 columns, so training covers that scale variation.
- **vertical resize** — the residual scale differs per well, so resize vertically with $U(0.8, 1.25)$ for multi-scale training (the physical mapping is restored at readout).

Evaluation used 5-fold CV grouped by well.

Ensemble and final submission

On top of the same input representation we built models with different backbones, heads, and resolutions, and selected them by **error decorrelation** rather than standalone accuracy. Single-model CV improved from

6.4 to 5.2 over two months, mostly from the coordinate system (chord, resolution, window) and the synthetic data.

The final submission is a fixed-weight linear blend of the four models below. Some slots average re-training runs of the identical configuration to reduce variance (CV values are after averaging).

Model	Input size (W×H)	Vertical resolution	Backbone	Head	CV (5-fold pooled)
1	512×384	0.5 ft/bin	EfficientNetV2-L	heatmap	5.19
2	512×256	1.0 ft/bin	EfficientNetV2-L	heatmap	5.20
3	512×512	0.5 ft/bin	ConvNeXt-S	heatmap	5.17
4	512×384	0.5 ft/bin	DINOv3 ViT-L/16	detection	5.48
Ensemble					**4.91**

Public LB 6.249 / private LB 6.411.

What didn't work (selected)

- Adding raw GR / learned GR matching / a GR denoiser as inputs — the model leaned on these "plausible-looking summary inputs" and specific folds collapsed (prior amplification). Showing the evidence as-is via the match heatmap was the safest.
- Explicit ridge decoding (Hough / DP), a forward-backward smoother on the particle filter, typewell-specialized models, self-training.
- Making the posterior multimodal (CE-only training, mixup) to rescue wrong-mode wells — the posterior never split; wrong-mode is a property of the evidence, not the loss. Investing in the "quality of the field" (frame, resolution, synthetic data) beat readout tricks.